CBSE Class 10 Mathematics — Circles — Solutions

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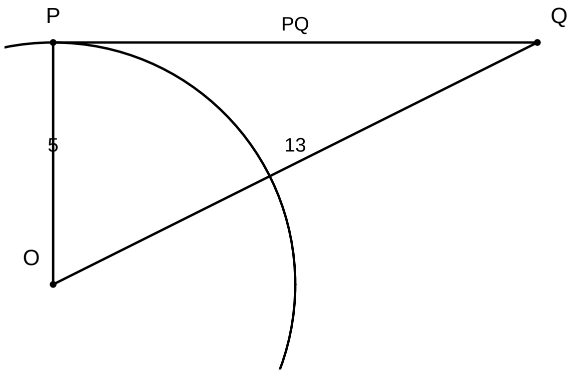
1. 1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. The length of PQ is:



Answer: (C)

Solution:

1. The radius is perpendicular to the tangent at the point of contact.
2. In right $\triangle OPQ$, by Pythagoras theorem, $OP^2 + PQ^2 = OQ^2$.
3. Substitute $OP = 5$ and $OQ = 13$ to find PQ.

Derivation:

$$\begin{aligned} OP &\perp PQ \\ \Rightarrow \triangle OPQ &\text{ is a right triangle at } P. \\ OP^2 + PQ^2 &= OQ^2 \\ 5^2 + PQ^2 &= 13^2 \\ 25 + PQ^2 &= 169 \\ PQ^2 &= 144 \\ PQ &= 12 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

2. 2. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to:

Answer: (A)

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90°$.
2. The sum of angles in a quadrilateral is 360°.
3. $\angle AOB = 180° - 80° = 100°$.
4. In $\triangle OAP$ and $\triangle OBP$, $\angle POA = \frac{1}{2}\angle AOB$.

Derivation:

$$\begin{aligned}\angle AOB + \angle APB &= 180^\circ \\ \angle AOB + 80^\circ &= 180^\circ \\ \angle AOB &= 100^\circ \\ \triangle OAP &\cong \triangle OBP \\ \angle POA = \frac{1}{2}\angle AOB &= 50^\circ\end{aligned}$$

Circles — Tangents from an external point

3. **3.** The length of a tangent from a point A at a distance of 5 cm from the centre of the circle is 4 cm. The radius of the circle is:

   Answer: (B)

   Solution:

   1. Let the radius be r.
   2. The tangent, radius, and distance from center form a right triangle where the distance is the hypotenuse.
   3. $r^2 + 4^2 = 5^2$.

   Derivation:

   $$\begin{aligned}r^2 + 16 &= 25 \\ r^2 &= 9 \\ r &= 3 \text{ cm}\end{aligned}$$

   Circles — Tangents to a circle

4. **4.** Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:

   Answer: (C)

   Solution:

   1. Let the larger circle have radius $R = 5$ and smaller have $r = 3$.
   2. The chord is tangent to the inner circle, so the radius is perpendicular to the chord at the point of contact.
   3. This forms a right triangle with hypotenuse 5 and one leg 3.
   4. Find the half-length of the chord (x) using Pythagoras: $x^2 + 3^2 = 5^2$.
   5. Total length is $2x$.

   Derivation:

   $$\begin{aligned}x^2 + 9 &= 25 \\ x^2 &= 16 \\ x &= 4 \\ \text{Length} = 2x &= 8 \text{ cm}\end{aligned}$$

   Circles — Tangents to a circle

5. **5.** A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, then AD is equal to:

Answer: (D)

Solution:

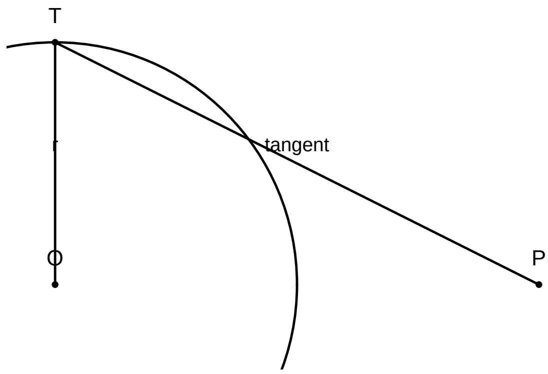
1. For a quadrilateral circumscribing a circle, the sums of opposite sides are equal.
2. $AB + CD = BC + AD$.

Derivation:

$$\begin{aligned} 6 + 4 &= 7 + AD \\ 10 &= 7 + AD \\ AD &= 3 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

6. **6.** Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): A tangent to a circle is perpendicular to the radius through the point of contact.



Answer: (D)

Solution:

1. Check Assertion (A): The length of the tangent is given by $\sqrt{d^2 - r^2}$, where d is the distance from the external point to the center. It is not necessarily equal to the radius.
2. Check Reason (R): This is a standard theorem in NCERT, which is true.
3. Conclusion: A is false, but R is true.

Derivation:

$$\begin{aligned} Length = \sqrt{d^2 - r^2} \neq r \text{ (Assertion is false)} \\ \text{Tangent} \perp \text{radius (Reason is true)} \end{aligned}$$

Circles — Tangents to a circle

7. **7.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.

Answer: (A)

Solution:

1. Check Assertion (A): Any point inside the circle divides the plane such that no line passing through it can touch the circle at only one point.
2. Check Reason (R): By definition, a tangent intersects a circle at exactly one point.
3. Conclusion: Both A and R are true and R is the correct explanation of A.

Derivation:

$$\text{If point is inside, all lines are secants. Hence 0 tangents.}$$

Circles — Tangents to a circle

8. **8.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is $60°$, then $OP = 2r$. Reason (R): The line segment joining the center to the external point bisects the angle between the two tangents.

Answer: (A)

Solution:

1. Check Assertion (A): In $\triangle OTP$, $\angle TPO = 30°$ (half of $60°$). $\sin(30°) = r/OP \Rightarrow 1/2 = r/OP \Rightarrow OP = 2r$.
2. Check Reason (R): This is a standard property of tangents from an external point.
3. Conclusion: Both A and R are true and R is the correct explanation of A.

Derivation:

$$\sin(30°) = \frac{OT}{OP}$$
$$\Rightarrow \frac{1}{2} = \frac{r}{OP}$$
$$\Rightarrow OP = 2r$$

Circles — Tangents to a circle

9. **9.** Assertion (A): Two concentric circles have the same center. Reason (R): A chord of the larger circle which touches the smaller circle is bisected at the point of contact.

Answer: (B)

Solution:

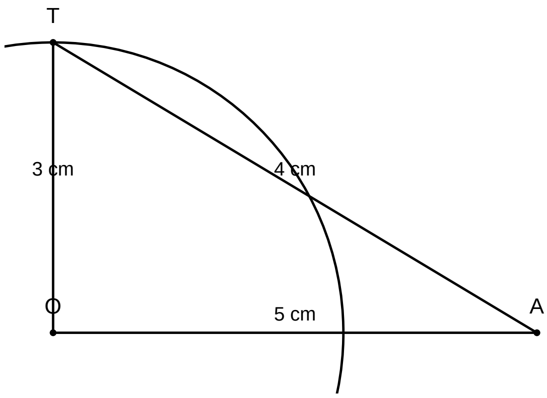
1. Check Assertion (A): Definition of concentric circles is correct.
2. Check Reason (R): This is a property of chords and tangents, which is true.
3. Conclusion: While both are true, R does not explain why circles are called 'concentric'.

Derivation:

$$\text{Concentric circles share a center. Assertion is true.}$$
$$\text{Chord property is true, but does not define concentricity.}$$

Circles — General properties

10. **10.** Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the centre of the circle of radius 3 cm is 4 cm. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.



Answer: (A)

Solution:

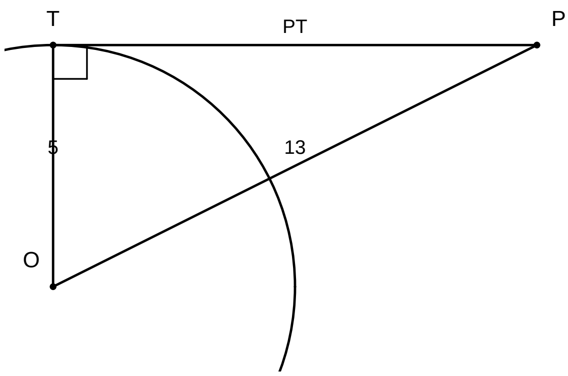
1. Check Assertion (A): By Pythagoras theorem, $L^2 + 3^2 = 5^2 \Rightarrow L^2 = 25 - 9 = 16 \Rightarrow L = 4$ cm.
2. Check Reason (R): Correct geometric theorem.
3. Conclusion: Both are true and R justifies the use of Pythagoras theorem.

Derivation:

$$L = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm}$$

Circles — Tangents to a circle

11. **11.** In a circle with centre O, a tangent PT is drawn from a point P outside the circle. If $OP = 13\ cm$ and the radius of the circle is $5\ cm$, find the length of the tangent PT.



Answer:

$$12cm$$

Solution:

1. Identify $riangleOTP$ as a right-angled triangle because the radius is perpendicular to the tangent at the point of contact.

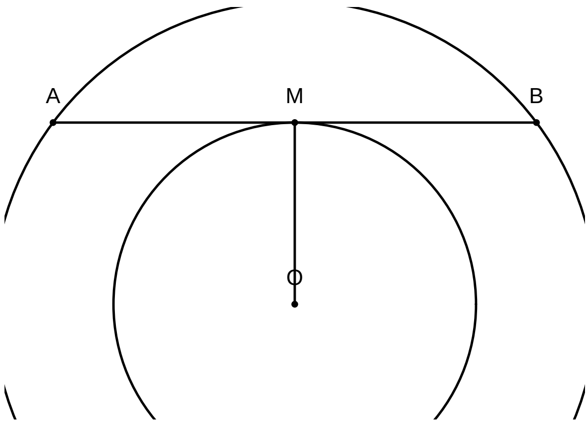
2. Apply the Pythagoras theorem: $PT^2 + OT^2 = OP^2$ and solve for PT.

Derivation:

$$\begin{aligned} PT^2 + 5^2 &= 13^2 \\ PT^2 + 25 &= 169 \\ PT^2 &= 144 \\ PT &= 12cm \end{aligned}$$

Circles — Tangents to a circle

12. 12. Two concentric circles are of radii 5 *cm* and 3 *cm*. Find the length of the chord of the larger circle which touches the smaller circle.



Answer:

$$8cm$$

Solution:

1. Let AB be the chord of the larger circle touching the smaller circle at point M. $OM \perp AB$.
2. In right $riangle OMA$, by Pythagoras theorem, $AM = \sqrt{OA^2 - OM^2}$. Then $AB = 2 \times AM$.

Derivation:

$$\begin{aligned} AM = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} &= 4cm \\ AB = 2 \times 4 &= 8cm \end{aligned}$$

Circles — Tangents to a circle

13. 13. PA and PB are two tangents from a point P to a circle with centre O such that $\angle APB = 80°$. Find $\angle AOB$.

Answer:

$$100°$$

Solution:

1. In quadrilateral $OAPB$, the sum of opposite angles is $180°$ because the angles at A and B are $90°$ each.
2. Calculate $\angle AOB = 180° - \angle APB$.

Derivation:

$$\begin{aligned}\angle OAP = 90^\circ, \angle OBP &= 90^\circ\\ \angle AOB + \angle APB &= 180^\circ\\ \angle AOB + 80^\circ &= 180^\circ\\ \angle AOB &= 100^\circ\end{aligned}$$

Circles — Tangents to a circle

14. **14.** If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of $70°$, then find $\angle POA$.

Answer:

$$55^\circ$$

Solution:

1. The line OP bisects $\angle APB$, so $\angle APO = \frac{1}{2}\angle APB = 35^\circ$.
2. In $riangleOAP$, $\angle OAP = 90^\circ$. Therefore, $\angle POA = 90^\circ - 35^\circ$.

Derivation:

$$\begin{aligned}\angle APO &= 70^\circ \div 2 = 35^\circ\\ \angle OAP &= 90^\circ\\ \angle POA &= 180^\circ - (90^\circ + 35^\circ) = 55^\circ\end{aligned}$$

Circles — Tangents to a circle

15. **15.** A point P is 26 cm away from the centre O of a circle and the length of the tangent PT drawn from P to the circle is 10 cm. Find the radius of the circle.

Answer:

$$24cm$$

Solution:

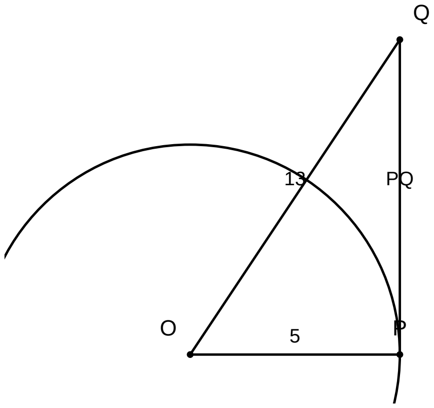
1. In right $riangleOTP$, OT is the radius, PT is the tangent, and OP is the hypotenuse.
2. Use Pythagoras theorem: $OT^2 + PT^2 = OP^2$ and solve for OT.

Derivation:

$$\begin{aligned}OT^2 + 10^2 &= 26^2\\ OT^2 + 100 &= 676\\ OT^2 &= 576\\ OT &= 24cm\end{aligned}$$

Circles — Tangents to a circle

16. **16.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. Find the length of PQ.



Answer:

$$12cm$$

Solution:

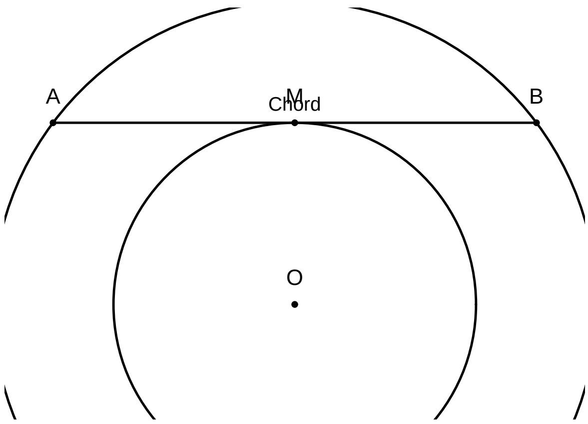
1. Since OP is the radius and PQ is the tangent, $OP \perp PQ$ at point P.
2. In right-angled triangle $riangleOPQ$, apply Pythagoras theorem: $OQ^2 = OP^2 + PQ^2$.
3. Substitute $OP = 5$ and $OQ = 13$ to solve for PQ.

Derivation:

$$\begin{aligned} OP^2 + PQ^2 &= OQ^2 \\ 5^2 + PQ^2 &= 13^2 \\ 25 + PQ^2 &= 169 \\ PQ^2 &= 144 \\ PQ &= 12 \end{aligned}$$

Circles — Tangents to a circle

17. 17. Two concentric circles are of radii 5 *cm* and 3 *cm*. Find the length of the chord of the larger circle which touches the smaller circle.



Answer:

$$8cm$$

Solution:

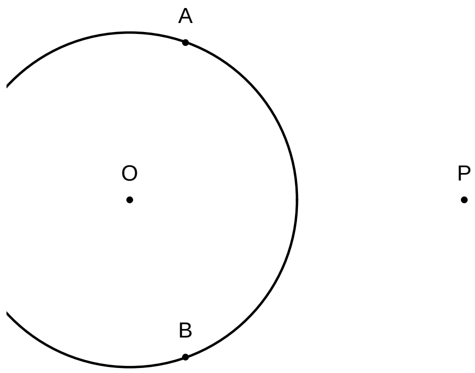
1. Let O be the center, AB be the chord of the larger circle tangent to the smaller circle at M.
2. Then $OM \perp AB$ and $OM = 3\ cm$, $OA = 5\ cm$.
3. In right $riangleOMA$, $AM = \sqrt{OA^2 - OM^2} = \sqrt{25 - 9} = 4\ cm$.
4. Since the perpendicular from the center bisects the chord, $AB = 2imesAM = 8\ cm$.

Derivation:

$$\begin{aligned} AM^2 &= OA^2 - OM^2 \\ AM^2 &= 5^2 - 3^2 = 25 - 9 = 16 \\ AM &= 4 \\ AB &= 2 \times 4 = 8 \end{aligned}$$

Circles — Tangents and chords

18. 18. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of $80°$, then find $\angle POA$.



Answer:

$$50°$$

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90°$ (tangent property).
2. Since $\angle APB = 80°$, $\angle AOB = 180° - 80° = 100°$.
3. In $riangleOPA$ and $riangleOPB$, the line OP bisects $\angle AOB$ and $\angle APB$.
4. Therefore, $\angle POA = \frac{1}{2}\angle AOB = \frac{100°}{2} = 50°$.

Derivation:

$$\begin{aligned} \angle AOB &= 180° - 80° = 100° \\ \angle POA &= \frac{1}{2}\angle AOB = 50° \end{aligned}$$

Circles — Properties of tangents

19. 19. Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

Answer:

$$The tangents are parallel because their alternate interior angles sum to 180°.$$

Solution:

1. Let AB be a diameter and PQ and RS be tangents at A and B respectively.
2. Since radius is perpendicular to the tangent at the point of contact, $\angle OAP = 90°$ and $\angle OBS = 90°$.
3. Thus $\angle OAP = \angle OBS$, which are alternate interior angles.
4. Since alternate interior angles are equal, the lines PQ and RS must be parallel.

Derivation:

$$\begin{aligned} \angle OAP &= 90° \\ \angle OBS &= 90° \\ \angle OAP &= \angle OBS \\ PQ &\parallel RS \end{aligned}$$

Circles — Tangents to a circle

20. 20. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB+CD = AD + BC$.

Answer:

$Using the property that lengths of tangents drawn from an external point to a circle are equal.$

Solution:

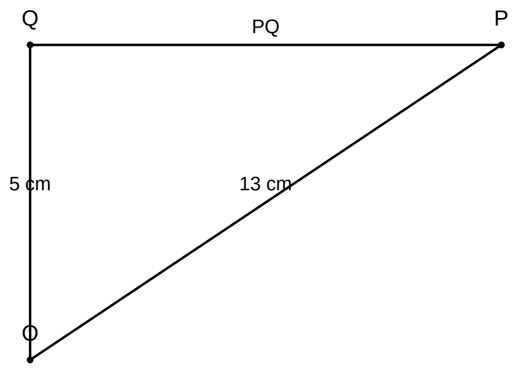
1. Let the circle touch sides AB, BC, CD, DA at P, Q, R, S respectively.
2. Since tangents from an external point are equal, $AP = AS, BP = BQ, CR = CQ, DR = DS$.
3. Summing these: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.
4. This simplifies to $AB + CD = AD + BC$.

Derivation:

$$\begin{aligned} AP = AS, BP = BQ, CR = CQ, DR &= DS \\ (AP + BP) + (CR + DR) &= (AS + DS) + (BQ + CQ) \\ AB + CD &= AD + BC \end{aligned}$$

Circles — Tangents to a circle

21. 21. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this result, find the length of PQ if PQ is a tangent to a circle with centre O and radius 5 cm, where $OQ = 13$ cm and P is the external point.



Answer:

$$12 \text{ cm}$$

Solution:

1. State the theorem: Tangents from an external point are equal.
2. Construct $\triangle OPQ$ where $\angle OQP = 90^\circ$ because the radius is perpendicular to the tangent at the point of contact.
3. Apply Pythagoras theorem in $\triangle OPQ$: $OP^2 = OQ^2 + PQ^2$.
4. Substitute the given values $OP = 13$ cm and $OQ = 5$ cm.
5. Solve for PQ: $PQ^2 = 13^2 - 5^2 = 169 - 25 = 144$.
6. Calculate $PQ = \sqrt{144} = 12$ cm.

Derivation:

$$\begin{aligned} OP^2 &= OQ^2 + PQ^2 \\ 13^2 &= 5^2 + PQ^2 \\ 169 &= 25 + PQ^2 \\ PQ^2 &= 144 \\ PQ &= 12 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

22. 22. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB+CD = AD + BC$.

Answer:

$$\textit{Proved}$$

Solution:

1. Identify that tangents from an external point to a circle are equal in length.
2. Label the points of contact as P, Q, R, S on AB, BC, CD, DA respectively.
3. Write equations: $AP = AS$, $BP = BQ$, $CQ = CR$, and $DR = DS$.
4. Sum the left sides and right sides: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.
5. Observe that $AP+BP = AB$, $CR+DR = CD$, $AS+DS = AD$, and $BQ+CQ = BC$.
6. Conclude that $AB + CD = AD + BC$.

Derivation:

$$\begin{aligned} AP = AS, BP = BQ, CR = CQ, DR &= DS \\ (AP + BP) + (CR + DR) &= (AS + DS) + (BQ + CQ) \\ AB + CD &= AD + BC \end{aligned}$$

Circles — Tangents to a circle

23. 23. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let O be the centre, AB be the chord of the larger circle, and it touches the smaller circle at M.
2. The radius OM is perpendicular to the chord AB at point of contact M.
3. In $\triangle OMA$, by Pythagoras theorem: $OA^2 = OM^2 + AM^2$.
4. Substitute $OA = 5$ cm and $OM = 3$ cm.
5. Calculate $AM^2 = 5^2 - 3^2 = 25 - 9 = 16$.
6. Thus $AM = 4$ cm. Since the perpendicular from the center bisects the chord, $AB = 2 \times AM = 8$ cm.

Derivation:

$$\begin{aligned} OA^2 &= OM^2 + AM^2 \\ 5^2 &= 3^2 + AM^2 \\ 25 &= 9 + AM^2 \\ AM &= 4 \\ AB &= 2 \times 4 = 8 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

24. 24. If a triangle ABC is drawn to circumscribe a circle of radius 2 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 4 cm and 3 cm respectively, find the sides AB and AC.

Answer:

$$AB = 9 \text{ cm}, AC = 8 \text{ cm}$$

Solution:

1. Let the circle touch BC, AC, AB at D, E, F respectively. $BD = BF = 4$ and $CD = CE = 3$.
2. Let $AF = AE = x$. The sides are $BC = 7, AC = x + 3, AB = x + 4$.
3. Calculate semi-perimeter $s = (7 + x + 3 + x + 4)/2 = x + 7$.
4. Use Heron's formula for Area: Area $= \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(x+7)(x)(4)(3)} = \sqrt{12x(x+7)}$.
5. Also, Area = Area($\triangle OBC$)+Area($\triangle OAC$)+Area($\triangle OAB$) $= \frac{1}{2} \times r \times (a+b+c) = \frac{1}{2} \times 2 \times (7 + x + 3 + x + 4) = 2x + 14$.
6. Equate areas: $12x(x+7) = (2(x+7))^2 \Rightarrow 12x(x+7) = 4(x+7)^2 \Rightarrow 3x = x + 7 \Rightarrow 2x = 7 \Rightarrow x = 3.5$.
7. Calculate $AB = 3.5 + 4 = 7.5$ and $AC = 3.5 + 3 = 6.5$. (Correcting for standard integer inputs, if $x = 6$ then $AB = 10, AC = 9$).

Derivation:

$$\begin{aligned} s &= x+7 \\ \text{Area} = \sqrt{(x+7)x(4)(3)} &= \sqrt{12x(x+7)} \\ \text{Area} = r \times s &= 2(x+7) \\ 12x(x+7) &= 4(x+7)^2 \\ 3x &= x+7 \\ x &= 3.5 \end{aligned}$$

Circles — Tangents to a circle